

AI_ENERGY_INFRA Intelligence Report

CONDITIONALLY_READY

report_id=report_intel_87210a61730c22b9_v4 version=4 view=PUBLIC

CURRENT STATE

US electric power consumption per capita (World Bank EG.USE.ELEC.KH.PC) ranged 12393.4-13392.7 kWh/capita over 2010-2022 (no AI-specific attribution; series ends before the 2023+ AI data-center buildout). Separately, live-acquired evidence now connects: (1) global data centers consumed ~415 TWh in 2024 (1.5% of world electricity), growing ~12%/yr since 2017, with IEA naming AI as a primary but not sole driver (AI_ATTRIBUTION=PARTIAL); (2) IEA forecasts (not observations) of ~945 TWh by 2030 and ~1200 TWh by 2035; (3) a Jevons-paradox dynamic in general cloud computing (Google total energy +3.7x 2016-2022 despite near-optimal PUE), AI_ATTRIBUTION=INDIRECT; (4) ~130 GW stuck in PJM's interconnection queue with ~\$3.5B estimated forgone savings, attributed jointly to data centers and electrification (AI_ATTRIBUTION=PARTIAL), framed as a generation/interconnection-queue shortage, not a transmission bottleneck. No source in this corpus gives a numeric AI-only (vs. general cloud/crypto) share of data center electricity demand. [v3 update] Real evidence has since been added for Europe (Ireland CSO: DC share of metered electricity 5%→22%, 2015→2024; EU EED Art.12 500kW+ reporting mandate, policy-only), the US (LBNL/DOE 2024 report: 176 TWh observed 2023 = 4.4% of US electricity, up from 58 TWh 2014, vs. 325-580 TWh by 2028 forecast -- kept as separate OBSERVED vs FORECAST claims), Korea (capital-region grid-connection bottleneck, 91.8% of contracted nationwide DC capacity concentrated in Seoul region per secondary reporting of National Assembly audit material, NOT independently verified against the primary document), and China (IEA ~100 TWh vs Goldman/Bloomberg ~200 TWh for 2024, both kept rather than resolved). Counterevidence acquisition also closed 2 of 3 remaining SEARCH_NOT_RUN directions with real findings (manufacturing/reshoring per Grid Strategies LLC; weather/peak-load per EIA/Ember, flagged as a peak-load-timescale finding distinct from annual TWh/yr demand growth); the DC-growth-without-power-problems region search ran but found no clear evidence either way (an honest negative, not a gap left unexamined). None of this new

evidence provides a numeric AI-only share of data-center electricity demand in any geography. [O-1B update] Hypothesis evaluation layer audited and extended: H3 and H4 corrected from SUPPORTED to PARTIALLY_SUPPORTED after applying a new deterministic Evidence Evaluation Contract (source tier / directness / AI-attribution-strength / temporal-type, plus same-origin independence checking) and 4 sufficiency guards; H1/H2/H7 reviewed and confirmed unchanged on quality grounds; H5/H6 reviewed, and a separate PRE-EXISTING discrepancy was found and disclosed (their canonical stored statuses, CONTESTED and WEAKENED, do not match what the unmodified classify_status() function would recompute from their own stored evidence arrays, which is REJECTED for both) -- left UNCHANGED as out of this phase's scope, not silently corrected. The H1/H3 relationship is now explicitly classified COMPLEMENTARY, not left as an implicit simultaneous-SUPPORTED artifact. [O-1C round-4 update] Hypothesis canonical write path re-evaluated: H1=SUPPORTED, H2=SUPPORTED, H3=SUPPORTED, H4=SUPPORTED, H5=WEAKENED, H6=WEAKENED, H7=SUPPORTED, each now carrying a canonical_status_diagnostics field (previous/legacy/quality-aware/final status, resolved/unresolved evidence, guard_trail) recorded in hypotheses.json. H1<->H3 both SUPPORTED is the expected, explicitly classified COMPLEMENTARY relationship (not a contradiction) per o1b_contradiction_resolution.json. New Geographic Applicability (o1c_geographic_applicability_result.json) and Temporal Applicability (o1c_temporal_applicability_result.json) sidecars were built this round, enforcing anti-overgeneralization rules (PJM != USA; US != Korea; Ireland != EU average; Korea contracted capacity != actual AI consumption) and OBSERVED-vs-FORECAST separation per claim. The 8-item alternative-explanations corpus (energy_demand_alternatives_result.json) was re-evaluated against the 14 canonical Claims now on file. A full synthesis rewrite (o1c_synthesis_intel_87210a61730c22b9.json) answers Te's 7 standing questions explicitly, with claim/hypothesis references, and marks UNKNOWN where evidence does not support an answer.

KEY SIGNAL

EVIDENCE_INSUFFICIENT

EVIDENCE_INSUFFICIENT

EVIDENCE STATUS

CONDITIONALLY_READY

MAJOR UNCERTAINTY

{'cause': 'ATTRIBUTION_UNCERTAINTY', 'detail': 'aggregate per-capita consumption cannot be decomposed by end-use sector from this indicator alone'}

WHAT TO WATCH

- ATTRIBUTION_GAP
- TEMPORAL_GAP
- COUNTEREVIDENCE_GAP
- POLICY_RESEARCH_GAP
- HYPOTHESIS_MODEL_QUALITY_WEIGHTING_GAP
- AI_ONLY_ATTRIBUTION_DISAGGREGATION_GAP

핵심 질문

AVAILABLE

- is AI data center growth driving US electricity demand increase?

현재 상태

AVAILABLE

- US electric power consumption per capita (World Bank EG.USE.ELEC.KH.PC) ranged 12393.4-13392.7 kWh/capita over 2010-2022 (no AI-specific attribution; series ends before the 2023+ AI data-center buildout). Separately, live-acquired evidence now connects: (1) global data centers consumed ~415 TWh in 2024 (1.5% of world electricity), growing ~12%/yr since 2017, with IEA naming AI as a primary but not sole driver (AI_ATTRIBUTION=PARTIAL); (2) IEA forecasts (not observations) of ~945 TWh by 2030 and ~1200 TWh by 2035; (3) a Jevons-paradox dynamic in general cloud computing (Google total energy +3.7x 2016-2022 despite near-optimal PUE), AI_ATTRIBUTION=INDIRECT; (4) ~130 GW stuck in PJM's interconnection queue with ~\$3.5B estimated forgone savings, attributed jointly to data centers and electrification (AI_ATTRIBUTION=PARTIAL), framed as a generation/interconnection-queue shortage, not a transmission bottleneck. No source in this corpus gives a numeric AI-only (vs. general cloud/crypto) share of data center electricity demand. [v3 update] Real evidence has since been added for Europe (Ireland CSO: DC share of metered electricity 5%->22%, 2015->2024; EU EED Art.12 500kW+ reporting mandate, policy-only), the US (LBNL/DOE 2024 report: 176 TWh observed 2023 = 4.4% of US electricity, up from 58 TWh 2014, vs. 325-580 TWh by 2028 forecast -- kept as separate OBSERVED vs FORECAST claims), Korea (capital-region grid-connection bottleneck, 91.8% of contracted nationwide DC capacity concentrated in Seoul region per secondary reporting of National Assembly audit material, NOT independently verified against the primary document), and China (IEA ~100 TWh vs Goldman/Bloomberg ~200 TWh for 2024, both kept rather than resolved). Counterevidence acquisition also closed 2 of 3 remaining SEARCH_NOT_RUN directions with real findings (manufacturing/reshoring per Grid Strategies LLC; weather/peak-load per EIA/Ember, flagged as a peak-load-timescale finding distinct from annual TWh/yr demand growth); the DC-growth-without-power-problems region search ran but found no clear evidence either way (an honest negative, not a gap left unexamined). None of this new evidence provides a numeric AI-only share of data-center electricity demand in any geography. [O-1B update] Hypothesis evaluation layer audited and extended: H3 and H4 corrected from SUPPORTED to PARTIALLY_SUPPORTED after applying a new deterministic Evidence Evaluation Contract (source tier / directness / AI-attribution-strength / temporal-type, plus same-origin independence checking) and 4 sufficiency guards; H1/H2/H7 reviewed and confirmed unchanged on quality grounds; H5/H6 reviewed, and a separate PRE-EXISTING discrepancy was found and disclosed (their canonical stored statuses, CONTESTED and WEAKENED, do not match what the unmodified classify_status() function would recompute

from their own stored evidence arrays, which is REJECTED for both) -- left UNCHANGED as out of this phase's scope, not silently corrected. The H1/H3 relationship is now explicitly classified COMPLEMENTARY, not left as an implicit simultaneous-SUPPORTED artifact. [O-1C round-4 update] Hypothesis canonical write path re-evaluated: H1=SUPPORTED, H2=SUPPORTED, H3=SUPPORTED, H4=SUPPORTED, H5=WEAKENED, H6=WEAKENED, H7=SUPPORTED, each now carrying a canonical_status_diagnostics field (previous/legacy/quality-aware/final status, resolved/unresolved evidence, guard_trail) recorded in hypotheses.json. H1<->H3 both SUPPORTED is the expected, explicitly classified COMPLEMENTARY relationship (not a contradiction) per o1b_contradiction_resolution.json. New Geographic Applicability (o1c_geographic_applicability_result.json) and Temporal Applicability (o1c_temporal_applicability_result.json) sidecars were built this round, enforcing anti-overgeneralization rules (PJM != USA; US != Korea; Ireland != EU average; Korea contracted capacity != actual AI consumption) and OBSERVED-vs-FORECAST separation per claim. The 8-item alternative-explanations corpus (energy_demand_alternatives_result.json) was re-evaluated against the 14 canonical Claims now on file. A full synthesis rewrite (o1c_synthesis_intel_87210a61730c22b9.json) answers Te's 7 standing questions explicitly, with claim/hypothesis references, and marks UNKNOWN where evidence does not support an answer.

핵심 주장

AVAILABLE

- 현재 확인되지 않았다. [OPEN](#)
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확인된 사실

INSUFFICIENT_EVIDENCE

Insufficient evidence in the current search scope.

아직 모르는 것 AVAILABLE

- {'gap_type': 'ATTRIBUTION_GAP', 'reason': 'partially addressed: IEA/GridLab evidence now connects AI/data-center growth to electricity demand at PARTIAL attribution (named as one driver among several -- general digital services, electrification -- with no numeric AI-only split given by any source); the earlier World Bank per-capita series still cannot be decomposed by end-use sector.'}
- {'gap_type': 'TEMPORAL_GAP', 'reason': 'series ends 2022, before the 2023+ AI data-center buildout commonly cited in AI_ENERGY_INFRA reporting -- cannot speak to the most recent period'}
- {'gap_type': 'COUNTEREVIDENCE_GAP', 'reason': 'partially addressed: live counterevidence acquisition (o0_live_counterevidence_supplement.json) found real evidence for efficiency-improvements and non-AI-cloud-growth directions via the Jevons paper; manufacturing/reshoring, EV/electrification, population growth, and weather/climate directions remain SEARCH_NOT_RUN or evidence-thin.'}
- {'gap_type': 'POLICY_RESEARCH_GAP', 'reason': 'carried over from M.5E-F -- still NOT_READY, unresolved'}
- {'gap_type': 'HYPOTHESIS_MODEL_QUALITY_WEIGHTING_GAP', 'reason': "intel/hypothesis/hypothesis_model.py classify_status() is purely evidence-COUNT-based with zero quality/independence weighting (if supporting_evidence and not contradicting_evidence: return SUPPORTED). As of this update it marks H1 (hyp_b0a4bbbc9b601728, 'AI data centers are a significant driver of electricity demand growth') and H3 (hyp_o0_h3_ai_energy_infra, 'recent demand growth results from multiple non-AI factors') BOTH as SUPPORTED simultaneously, even though they represent competing explanations -- because each merely has >=1 supporting claim and 0 directly-contradicting claims in the model's narrow count-based sense. This is a disclosed structural limitation of the current hypothesis model, not a data error and not silently corrected here: classify_status() was intentionally left unmodified this phase per explicit instruction, since rewriting its logic would be an unscoped architecture change. Any report or synthesis reading H1/H3 status must present both as SUPPORTED-under-this-model while explicitly flagging that the model cannot currently adjudicate between competing hypotheses by evidence quality, independence, or directness of contradiction. [O-1B update] PARTIALLY RESOLVED: a new, additive Evidence Evaluation Contract (intel/hypothesis/evidence_evaluation.py) and deterministic sufficiency guards (Forecast/AI-Attribution/Derivative-Evidence/Industry-Self-Report) were added in

intel/hypothesis/hypothesis_model.py.evaluate_hypothesis_sufficiency(), and applied to the real AI_ENERGY_INFRA corpus. Result: H3 (hyp_o0_h3_ai_energy_infra) is downgraded from SUPPORTED to PARTIALLY_SUPPORTED (its 4 supporting_evidence references do not resolve to any evaluable Claim record -- SOURCE_TIER/DIRECTNESS both UNKNOWN across the board), and H4 is likewise downgraded to PARTIALLY_SUPPORTED for the same reason. H1 remains SUPPORTED on its own evidentiary merits (passes every guard, with real PRIMARY_RESEARCH-tier, AI_PARTIAL-attribution evidence). The H1/H3 relationship is now explicitly classified as COMPLEMENTARY (not MUTUALLY_EXCLUSIVE) in intel/hypothesis/o1b_contradiction_resolution.json, with full reasoning. NOT fully resolved: classify_status() itself (the plain count-based function) is left unmodified and is still what upsert_hypothesis() uses on every future evidence append -- a hypothesis corrected by this new layer will silently revert to its count-based status on its next evidence-array update unless evaluate_hypothesis_sufficiency() is re-applied. That residual architectural gap is why this is PARTIALLY_RESOLVED, not RESOLVED. [O-1C round-4 update] Re-evaluated against Te's section 24 six conditions: (1) canonical write path quality-aware = YES (evaluate_hypothesis_sufficiency() is the live write path for this round's H1-H7 re-evaluation); (2) unresolved-evidence guard working = YES (verified operating during the round-2/3 canonical migration); (3) migrated evidence canonical resolution complete = PARTIALLY (most key GridLab/manufacturing-reshoring/EV/climate evidence migrated to canonical Claims claim_a11c5cb4c7c4c2d1/claim_f5842e6162c8fefc/claim_d641fb1790c3ef53, but a few references -- e.g. the IEA executive-summary citation and the Ireland CSO reference underlying H4 -- remain pointed at non-Claim provenance strings rather than resolvable Claim records, and population-growth/industrial-policy alternatives remain under-researched); (4) geography applicability reflected = YES (o1c_geographic_applicability_result.json, new this round); (5) temporal applicability reflected = YES (o1c_temporal_applicability_result.json, new this round); (6) status-reversion test passing = YES. Because condition (3) is only PARTIALLY satisfied, this gap stays PARTIALLY_RESOLVED rather than RESOLVED -- classify_status() itself is also still unmodified, which remains the deeper residual architectural gap.", 'status': 'PARTIALLY_RESOLVED'}

- {'gap_type': 'AI_ONLY_ATTRIBUTION_DISAGGREGATION_GAP', 'reason': 'Confirmed again this round across all 14 canonical claims and all 7 geographic-applicability regions: no source (IEA, LBNL/DOE, Ireland CSO, Carbon Brief/Oxford on China, Korea National Assembly secondary reporting) gives a numeric AI-only (vs general cloud/crypto/colocation) TWh or percentage split of data-center electricity demand, in any geography. This is distinct from the pre-existing ATTRIBUTION_GAP (which concerns the older World Bank per-capita series) and is recorded separately because it now rests on explicit cross-referencing of the full

claims/hypothesis/geographic/temporal/alternative-explanations evidence base rather than a single series.', 'status': 'UNRESOLVED'}

- {'cause': 'ATTRIBUTION_UNCERTAINTY', 'detail': 'aggregate per-capita consumption cannot be decomposed by end-use sector from this indicator alone'}
- {'cause': 'TEMPORAL_UNCERTAINTY', 'detail': 'no data points after 2022 in this series'}
- {'cause': 'METHODOLOGY_UNCERTAINTY', 'detail': "Hypothesis statuses (H1-H7) are produced by a deterministic but count-only classifier; a status of SUPPORTED reflects 'at least one supporting claim, no claim coded as directly contradicting it' and must not be read as 'this hypothesis has been validated against competing explanations.' The H1/H3 simultaneous-SUPPORTED outcome is the clearest symptom."}
- {'cause': 'RESIDUAL_METHODOLOGY_UNCERTAINTY', 'detail': 'O-1B added a deterministic Evidence Evaluation Contract and 4 sufficiency guards, which corrected H3/H4 from SUPPORTED to PARTIALLY_SUPPORTED and explicitly classified the H1/H3 relationship as COMPLEMENTARY -- but this does not mean AI's specific electricity consumption share is now measured: it still is not measured anywhere in this corpus, by any source, in any geography. Cross-country comparability (US/Ireland/Korea/China use different definitions of "data center" and different measurement regimes), the US-vs-Korea/Europe/China temporal-range gaps (LBNL 2014-2023 vs. current 2024-2026 AI-diffusion-era dynamics), and reliance on corporate self-reports for any future efficiency claims all remain real, unresolved uncertainty. [O-1C round-4 update] Geographic and temporal applicability sidecars now make the region/period scope of every piece of resolved H1-H7 evidence explicit and prevent silent generalization (PJM!=USA, US!=Korea, Ireland!=EU-average, Korea-contracted-capacity!=actual-AI-consumption), but they do not supply new measurement: AI's specific electricity-consumption share remains unmeasured in every geography in this corpus.'}

통계

AVAILABLE

Statistical series: IEA_GLOBAL_DC_ELECTRICITY_TWH_FORECAST (GLOBAL, 2030-2035) -- chart type LINE_CHART_NO_FORECAST, no forecast line, no AI-attribution overlay.

- | | |
|-----------|--|
| Indicator | IEA_GLOBAL_DC_ELECTRICITY_TWH_FORECAST |
| Geography | GLOBAL |
| Period | 2030-2035 |
| Source | IEA_ENERGY_AND_AI_EXEC_SUMMARY |
| Trend | INSUFFICIENT_SERIES |

실제 관측값 범위를 그대로 사용 -- 추세를 과장하는 축 조정 없음

Statistical series: EG.USE.ELEC.KH.PC (USA, 2010-2022) -- chart type LINE_CHART_NO_FORECAST, no forecast line, no AI-attribution overlay.

- | | |
|-----------|-------------------|
| Indicator | EG.USE.ELEC.KH.PC |
| Geography | USA |
| Period | 2010-2022 |
| Source | src_worldbank_api |
| Trend | NO_CLEAR_SIGNAL |

실제 관측값 범위를 그대로 사용 -- 추세를 과장하는 축 조정 없음

Statistical series: IEA_GLOBAL_DC_ELECTRICITY_TWH_OBSERVED (GLOBAL, 2024-2024) -- chart type TABLE, no forecast line, no AI-attribution overlay.

- | | |
|-----------|--|
| Indicator | IEA_GLOBAL_DC_ELECTRICITY_TWH_OBSERVED |
| Geography | GLOBAL |
| Period | 2024-2024 |
| Source | IEA_ENERGY_AND_AI_EXEC_SUMMARY |
| Trend | INSUFFICIENT_SERIES |

실제 관측값 범위를 그대로 사용 -- 추세를 과장하는 축 조정 없음

연구 근거

INSUFFICIENT_EVIDENCE

Insufficient evidence in the current search scope.

정책/규제

AVAILABLE

- {'policy': 'EU_EED_2023_1791_ART12', 'status': 'POLICY_ANNOUNCEMENT_ONLY', 'detail': '500kW+ data center energy reporting mandate effective 2024-09; no observed-effect data found yet'}
-

반증

AVAILABLE

- {'direction': 'manufacturing/reshoring', 'status': 'EVIDENCE_FOUND', 'source': 'Grid Strategies LLC reports'}
 - {'direction': 'weather/climate peak-load', 'status': 'EVIDENCE_FOUND', 'source': 'EIA/Ember peak-load spikes, ERCOT/Eastern Interconnection', 'note': 'flagged as a PEAK-LOAD timescale finding, distinct from annual TWh/yr trend claims -- not merged with them'}
 - {'direction': 'DC growth without power problems (region search)', 'status': 'SEARCH_RUN_NO_CLEAR_EVIDENCE', 'note': 'honest negative -- search was run, no clear supporting evidence found either way'}
-

대안 설명

AVAILABLE

- GENERAL_CLOUD_COMPUTING_GROWTH
- MANUFACTURING_RESHORING
- EV_ELECTRIFICATION
- WEATHER_PEAK_LOAD
- POPULATION_GROWTH
- DC_GROWTH_WITHOUT_POWER_PROBLEMS_REGIONS
- CLIMATE_POLICY_DECARBONIZATION_LOAD_SHIFT
- CRYPTO_MINING_LOAD

불확실성

AVAILABLE

- {'cause': 'ATTRIBUTION_UNCERTAINTY', 'detail': 'aggregate per-capita consumption cannot be decomposed by end-use sector from this indicator alone'}
- {'cause': 'TEMPORAL_UNCERTAINTY', 'detail': 'no data points after 2022 in this series'}
- {'cause': 'METHODOLOGY_UNCERTAINTY', 'detail': '"Hypothesis statuses (H1-H7) are produced by a deterministic but count-only classifier; a status of SUPPORTED reflects 'at least one supporting claim, no claim coded as directly contradicting it' and must not be read as 'this hypothesis has been validated against competing explanations.' The H1/H3 simultaneous-SUPPORTED outcome is the clearest symptom.'"}}
- {'cause': 'RESIDUAL_METHODOLOGY_UNCERTAINTY', 'detail': 'O-1B added a deterministic Evidence Evaluation Contract and 4 sufficiency guards, which corrected H3/H4 from SUPPORTED to PARTIALLY_SUPPORTED and explicitly classified the H1/H3 relationship as COMPLEMENTARY -- but this does not mean AI's specific electricity consumption share is now measured: it still is not measured anywhere in this corpus, by any source, in any geography. Cross-country comparability (US/Ireland/Korea/China use different definitions of "data center" and different measurement regimes), the US-vs-Korea/Europe/China temporal-range gaps (LBNL 2014-2023 vs. current 2024-2026 AI-diffusion-era dynamics), and reliance on corporate self-reports for any future efficiency claims all remain real, unresolved uncertainty. [O-1C round-4 update] Geographic and temporal applicability sidecars now make the region/period scope of every piece of resolved H1-H7 evidence explicit and prevent silent generalization (PJM!=USA, US!=Korea, Ireland!=EU-average, Korea-contracted-capacity!=actual-AI-consumption), but they do not supply new measurement: AI's specific electricity-consumption share remains unmeasured in every geography in this corpus.'}

지리적 맥락

AVAILABLE

- {'geography': 'USA', 'basis': 'statistic geography field (World Bank country code), not inferred from publisher or language', 'scope': 'event_country=UNKNOWN (per M.5E-F geography model discipline)'}
- {'geography': 'KR', 'basis': "source_registry_verified_publisher (connected event's documents)", 'scope': 'publisher_country only for event evt_a6a4027d9878143d; event_country stays UNKNOWN (no inference from publisher to event location)'}
- {'geography': 'KR', 'basis': 'Korea capital-region grid-connection bottleneck claim (secondary reporting of National Assembly audit material, via Rep. 이소영; KEPCO 9th T&D plan)', 'scope': 'DERIVATIVE_REPORTING/NEWS-SECONDARY, not promoted to POLICY_RESEARCH -- primary 국정감사 document not independently fetched'}
- {'geography': 'IE', 'basis': 'Ireland CSO official statistics (national statistical office, PRIMARY_SOURCE)', 'scope': "data centers = 22% of Ireland's metered electricity 2024, up from 5% in 2015; AI_ATTRIBUTION=INDIRECT"}
- {'geography': 'CN', 'basis': 'IEA vs Goldman Sachs/Bloomberg estimates via Carbon Brief / Oxford Institute for Energy Studies (DERIVATIVE_REPORTING/ACADEMIC_ANALYSIS)', 'scope': '~100 TWh (IEA) vs ~200 TWh (Goldman/Bloomberg) 2024, both kept side by side; East Data West Computing (东数西算) program described, not independently verified as primary source'}
- {'geography': 'MULTI_REGION_APPLICABILITY_MATRIX', 'basis': 'intel/geographic_evidence/o1c_geographic_applicability_result.json (new this round)', 'scope': "Per-hypothesis (H1-H7), per-region (USA/PJM/IRELAND/EU/CHINA/KOREA/SEOUL_METRO) MATCHED/PARTIAL_MATCH/OUT_OF_SCOPE/UNKNOWN classification, enforcing anti-overgeneralization rules: PJM evidence never promoted to USA-wide MATCHED; US evidence never applied to Korea or vice versa; Ireland's DC-electricity-share never used as an EU-wide average; Korea's contracted DC capacity never converted into actual AI consumption."}

시간적 맥락

AVAILABLE

- {'period_start': '2010', 'period_end': '2022', 'source': 'series_af37f1207e7ff042'}
- {'period_start': '2026-09-29T10:00:04+09:00', 'period_end': '2026-09-29T10:00:04+09:00', 'source': 'evt_a6a4027d9878143d', 'note': '1 real event, 2 documents reporting it -- document count != event count'}
- {'period_start': '2014', 'period_end': '2035', 'source': 'intel/geographic_evidence/o1c_temporal_applicability_result.json (new this round)', 'note': 'Per-hypothesis OBSERVED-vs-FORECAST classification (CURRENT_MATCH/HISTORICAL_CONTEXT/FORECAST_ONLY/PARTIAL_TEMPORAL_MATCH/OUT_OF_RANGE/UNKNOWN) against a 2024-2025 now_baseline; forecast horizons (2028/2030/2035) are explicitly never read as current.'}

근거 지도

AVAILABLE

- {'node_type': 'STATISTIC', 'id': 'series_67a90dc183bd74a1'}
- {'node_type': 'STATISTIC', 'id': 'series_af37f1207e7ff042'}
- {'node_type': 'STATISTIC', 'id': 'series_d018d1290acd5f20'}
- {'node_type': 'EVIDENCE_RELATION', 'id': 'rel_290e45f3e25fcb78'}
- {'node_type': 'EVIDENCE_RELATION', 'id': 'rel_5561f393ab96d55f'}
- {'node_type': 'EVIDENCE_RELATION', 'id': 'rel_8396e8c80f83a5b4'}
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- {'node_type': 'EVIDENCE_RELATION', 'id': 'rel_0cf98d7daad51b15'}
- {'node_type': 'CLAIM', 'id': 'claim_1d9458f4e7e2cf69'}
- {'node_type': 'CLAIM', 'id': 'claim_3ab07077d739a9a4'}
- {'node_type': 'CLAIM', 'id': 'claim_5787125ab5f0110b'}
- {'node_type': 'CLAIM', 'id': 'claim_7f1bc452c4ffc128'}
- {'node_type': 'CLAIM', 'id': 'claim_931b08c7ebfc9604'}
- {'node_type': 'CLAIM', 'id': 'claim_afe484e45964c605'}
- {'node_type': 'CLAIM', 'id': 'claim_afe7cc7b19317ee0'}
- {'node_type': 'CLAIM', 'id': 'claim_d170f52f567053e9'}
- {'node_type': 'CLAIM', 'id': 'claim_o1_korea_capital_region_dc_demand'}
- {'node_type': 'CLAIM', 'id': 'claim_o1_lbnl_us_dc_2023'}
- {'node_type': 'INTELLIGENCE_OBJECT', 'id': 'intel_87210a61730c22b9'}

출처

AVAILABLE

- {'source_id': 'event:evt_a6a4027d9878143d', 'note': 'non-document provenance reference'}
- {'source_id': 'https://api.worldbank.org/v2/country/USA/indicator/EG.USE.ELEC.KH.PC?date=2010:2022&format=json&per_page=100', 'note': 'non-document provenance reference'}
- {'source_id': 'https://arxiv.org/html/2411.11540v1', 'note': 'non-document provenance reference'}
- {'source_id': 'https://gridlab.org/interconnection-bottlenecks-cost-pjm-customers-3-5-billion/', 'note': 'non-document provenance reference'}
- {'source_id': 'https://www.iea.org/reports/energy-and-ai/executive-summary', 'note': 'non-document provenance reference'}

Sources

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METAXIS Intelligence Observatory -- structured evidence, not editorial conclusions.